

Answers

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|-------|-------|
| 1. D | 11. D |
| 2. C | 12. A |
| 3. D | 13. A |
| 4. A | 14. C |
| 5. C | 15. D |
| 6. B | 16. C |
| 7. B | 17. A |
| 8. A | 18. C |
| 9. D | 19. C |
| 10. B | 20. C |

21. Supervised learning is a type of machine learning where user has labeled data input inputted in the dependencies in order to find the output. There is teacher that tells which data has to be inputted and which data is not. Example includes: decision trees,

22. A model M1 gives 99% of accuracy for training dataset and it gives 30% of accuracy for testing dataset.

Comment on model M1:

- Training dataset has proper input values
- Testing dataset has missing input values.

23. Regression analysis is the type of regression model where user analyze of which input values should the regression takes and which values should be deleted or that are irrelevant for regression.

24. In Machine Learning there are many challenges including:

- Finding the correct data to do machine learning on.

Because not only that data are very complex, but we need to find the data which can represent the field which we need to study.

- Optimizing data: After finding relevant data, user needs to optimize them in order for the machine's performance better, faster and more accurate.

25. Two major applications of data science:

- Weather app: It uses data of winds, airs, rains, humidity to accurately predicts the weather like when it rains, average temperature, ...

- Language translation app: Ex: Googletranslate, Apple translate, uses machine learning to translate real time language input source to desired translated output language.

26. Distinguish between the terms: Structured and Semi-Structured data.

- Structured: All of the category of data or input source are categorized in the same type such as columns and rows so that they are easy to read and analyze.
- Semi-structured data: Most if not some are categorized in the same type and the remaining type of data is unknown or unclassified.

27. K-means algorithm is the name of the numbers of clusters.

28. Data pre-processing is:

- Loading of the input dataset
- Input requires module
- We need a technique to model to be able to classify the data.

29. Normalization requires in K-NN because we need to make the data readable and understandable to humans and to do that K-NN helps with changing type of data.

27. We find out an outlier in data by using machine learning algorithm to identify which data are the same and which are not the same.

30. Continuous attributes are discretized by finding which data are continuous data and then finding where it continues and where it stops.

32. Hyperparameters of machine learning:

- Name
- Date
- Integers
- Types

33. Gradient descent is

34. Comment on performance:

- First one is fast
- Second one is slow
- Third one is fastest.

35. Radial Basis function kernel functions in SVM.